

Question # 6 of 10 (Start time: 01:32:44 PM, 08 July 2025)

If A is an $n \times n$ matrix, then

Select the correct option

☒ $\det A^T = \det A$

☐ $\det A^T = -\det A$

Question # 5 of 10 (**Start time: 01:32:29 PM, 08 July 2025**)

If A is not invertible matrix, then

Select the correct option

☒	$\text{Det}(A)=0$
☐	$\text{Det}(A)=-1$
☐	$\det(A) \neq 0$
☐	$\text{Det}(A)=1$

Question # 4 of 10 (Start time: 01:32:13 PM, 08 July 2025)

Jacobi's method assumes that the diagonal entries of an invertible matrix A are all

Select the correct option

☒	non-zero
☐	zero

Question # 2 of 10 (Start time: 01:31:34 PM, 08 July 2025)

The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its main diagonal.

Select the correct option

FALSE



TRUE



Question # 9 of 10 (Start time: 01:34:50 PM, 08 July 2025)

Consider an LU-factorization, the matrix L is _____ and is called a unit lower triangular matrix.

Select the correct option

☐	Invertible
☐	zero
☒	Identity
☐	Singular

Question # 8 of 10 (Start time: 01:34:00 PM, 08 July 2025)

If two rows of a square matrix A are interchanged to produce B , then

Select the correct option

☐	$\det B = \det A$
☐	$\det B = \det A$
☐	none of the above
☒	$\det B = -\det A$

Question # 10 of 10 (Start time: 01:35:08 PM, 08 July 2025)

A decomposition of a matrix as a product of two or more matrices is called the matrix _____.

Select the correct option

☐	Composition
☐	Multiplication
☒	Factorization
☐	Transformation

Question # 3 of 10 (Start time: 01:31:52 PM, 08 July 2025)

Total Marks: 1

If the coefficient matrix is large and with a high proportion of zero entries, then which of the following works quickly in favor of solving a linear systems for saving less computer memory?

Select the correct option

☐	Reduced row-echelon method
☐	Row-echelon method
☒	Iterative method
☐	Direct method

Question # 1 of 10 (Start time: 01:31:15 PM, 08 July 2025)

If one row of a square matrix A is added to another row, then the resulting determinant will

Select the correct option

☐	None of the above
☐	Differ
☒	Remain same
☐	Equal

Question # 7 of 10 (Start time: 01:33:23 PM, 08 July 2025)

Total Marks: 1

If a system of equations is solved using the Jacobi's method , then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?

Select the correct option

☐	diagonal must be zero.
☐	All of its entries above the diagonal must be zero.
☐	All of its entries below the diagonal must be zero.
☒	All of its entries on the diagonal must be zero.